Motor Full-Load Current Chart

Full-load current by horsepower for three-phase and single-phase motors, as used to size motor circuit conductors and protection.

For sizing a motor circuit, the full-load current comes from the code table for the motor's horsepower and voltage, not from its nameplate. A 50 hp three-phase motor is taken as 65 A at 460 V and 130 A at 230 V. The table figure is used for the conductors, the short-circuit protection and the disconnect; the nameplate current is used only for the overload protection.

Three-phase induction motors

kW (hp)	208 V	230 V	460 V	575 V
0.373 (1/2)	2.4	2.2	1.1	0.9
0.559 (3/4)	3.5	3.2	1.6	1.3
0.746 (1)	4.6	4.2	2.1	1.7
1.12 (1-1/2)	6.6	6	3	2.4
1.49 (2)	7.5	6.8	3.4	2.7
2.24 (3)	10.6	9.6	4.8	3.9
3.73 (5)	16.7	15.2	7.6	6.1
5.59 (7-1/2)	24.2	22	11	9
7.46 (10)	30.8	28	14	11
11.2 (15)	46.2	42	21	17
14.9 (20)	59.4	54	27	22
18.6 (25)	74.8	68	34	27
22.4 (30)	88	80	40	32
29.8 (40)	114	104	52	41
37.3 (50)	143	130	65	52
44.7 (60)	169	154	77	62
55.9 (75)	211	192	96	77
74.6 (100)	273	248	124	99
93.2 (125)	343	312	156	125
112 (150)	396	360	180	144
149 (200)	528	480	240	192
186 (250)	—	—	302	242
224 (300)	—	—	361	289
261 (350)	—	—	414	336
298 (400)	—	—	477	382
336 (450)	—	—	515	412
373 (500)	—	—	590	472

Full-load current in amperes. The table is defined by horsepower; the kW figure is the same rating converted at 0.7457 kW per hp, and is not an IEC standard motor size.

Single-phase motors

kW (hp)	115 V	230 V
0.124 (1/6)	4.4	2.2
0.186 (1/4)	5.8	2.9
0.249 (1/3)	7.2	3.6
0.373 (1/2)	9.8	4.9
0.559 (3/4)	13.8	6.9
0.746 (1)	16	8
1.12 (1-1/2)	20	10
1.49 (2)	24	12
2.24 (3)	34	17
3.73 (5)	56	28
5.59 (7-1/2)	80	40
7.46 (10)	100	50

Full-load current in amperes. The kW figure is the horsepower rating converted at 0.7457 kW per hp.

Where the values come from

The three-phase currents correspond to NEC Table 430.250 and the single-phase currents to Table 430.248. They are reproduced here for reference and are not computed by Voltfield. The figures are the same pattern throughout: the 230 V current is twice the 460 V current, and the 208 V current is ten percent higher than the 230 V one. Check any value you intend to rely on against the edition of the code your jurisdiction has adopted.

These are code values, not Voltfield's. NEC and National Electrical Code are trademarks of the National Fire Protection Association, which publishes the code. Voltfield is not affiliated with it. The published code, in the edition adopted where you are, is the authority.

What this chart does not tell you

These are code values for sizing, and they are deliberately conservative. An actual motor, particularly a high-efficiency one, usually draws less. The tables do not cover low-speed motors of 1,200 rpm or less, high-torque motors or multispeed motors, which can draw more and are sized from the nameplate. They do not give locked-rotor current, and they do not size the overload device.

This is a reference, not a design. It is here to give you the right order of magnitude and a figure to check against. It does not replace the adopted code or a calculation by a qualified engineer.

From voltfield.org/charts/motor-full-load-current-chart.html, where the notes on reading this chart and a calculator for other cases are. Reviewed November 11, 2026. Free to print, copy and share.